Gangatharan Idayachandiran

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EDUCATION

Northeastern University, Khoury College of Computer Sciences, Boston, MA — Sep 2025 – Expected May 2027
Master of Science in Computer Science | GPA: 3.60/4.00
Coursework: Data Structures & Algorithms, Database Management Systems, Cloud Computing, Programming Design Paradigms

Vellore Institute of Technology, Chennai, India — Sep 2021 – Apr 2025
Bachelor of Technology, Computer Science and Engineering (AI & ML)

EXPERIENCE

Software Engineering Intern, TensorIoT, Boston, MA — Jun 2026 – Present
- Architected an edge-to-cloud AI pipeline for a retail security platform (IoT Core/MQTT, Bedrock, Polly, Lambda, CDK) and cut camera-to-spoken-response latency from 8s to 4s toward a 2s SLA; led design reviews with client executives.
- Built the full-stack multi-tenant operator console and customer dashboard (React, Lambda, DynamoDB) with MFA, role-based access control, branded onboarding emails, and an embedded third-party payment integration.
- Added per-stage latency and cost tracking, CloudWatch alerting, and DynamoDB and Redshift audit logging for multiple partners sharing the database; fixed CORS and error-handling issues and owned the technical decisions.
- Set up a CI pipeline for a team using AI-assisted coding workflows that scans commits for hardcoded secrets (TruffleHog, Gitleaks) and runs Playwright smoke tests, catching issues before deployment.

Web Development Lead, NU Launch Labs, Boston, MA — Sep 2026 – Present
- Lead a small team building the Launch Labs website for ~8,000 CS and research students, owning the technical specs, data model (teams, events, people, RSVPs), and phased delivery on Next.js, TypeScript, and Sanity CMS.
- Build the responsive, WCAG 2.1 AA front end with an event calendar and an RSVP flow (capacity limits, duplicate prevention) backed by Postgres, plus an email sign-up with client- and server-side validation.

Research Intern, Centre for Cyber Physical Systems, VIT, Chennai, India — Jun – Jul 2024
- Designed a CNN-Transformer hybrid in PyTorch for mammogram classification on CBIS-DDSM (2,620 images), selecting its components by benchmarking pretrained CNN and transformer models.
- Owned the full pipeline from preprocessing and augmentation through training, evaluation, and visualization, reaching 88.69% accuracy and 0.95 AUC on CBIS-DDSM and 92.86% on the independent MIAS dataset; added SHAP explainability.
- Co-authored and revised the manuscript, published in *The European Physical Journal Special Topics* (Springer, April 2025).

PROJECTS

Ledger | Spring Boot, React Native, PostgreSQL — Jun 2026 – Jul 2026
- Built a full-stack bill-splitting mobile app (React Native) backed by a Spring Boot REST API and PostgreSQL, deployed on Render, letting users create groups, log shared expenses, and settle balances across devices.
- Designed the schema and endpoints to compute balances server-side so every device sees the same settlement state; secured access with JWT auth, refresh-token rotation, and hashed credentials, tested end to end on real devices.

Real Estate Management System | Python, Django, MySQL — Sep 2025 – Dec 2025
- Built a property management web app with the data layer in raw SQL (PyMySQL): a 3NF schema of 10+ tables, with stored procedures, triggers, and user-defined functions enforcing business rules at the query level instead of the Django ORM.
- Added role-based access for Admin, Agent, and Client roles, CSRF protection, multi-filter property search, and an analytics dashboard; deployed on Railway with GitHub Actions CI/CD.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Libraries: Spring Boot, Django, FastAPI, React, React Native, PyTorch, Scikit-learn, Pandas, NumPy
Cloud (AWS**):** CDK, Lambda, IoT Core, API Gateway, Bedrock, Polly, SageMaker, CloudWatch, S3, IAM
Databases: PostgreSQL, MySQL, DynamoDB, Redshift
Testing & CI/CD: JUnit 5, Playwright, Postman, GitHub Actions, TruffleHog, Gitleaks
Tools: Git, Docker, Cursor, Claude Code
Certifications: AWS Certified Cloud Practitioner; Azure Administrator Associate (AZ-104)